

2.3 Trigonometry

Pure Mathematics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

At this level, double-angle identities $\sin 2x = 2 \sin x \cos x$ and $\cos 2x = 2 \cos^2 x - 1$ help solve equations and integrate. State the full interval after algebra and test solutions excluded by any division by $\sin x$ or $\cos x$.

Worked example: follow the reasoning

$\cos 2x = 1 - 2 \sin^2 x$ turns $2 \sin^2 x = 1$ into $\cos 2x = 0$, then interval solutions are selected.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Solve $\sin 2x = 0$ for $0 \leq x < 2\pi$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$2x = n\pi$ [1]; $x = n\pi/2$ [1]; $x = 0, \pi/2, \pi, 3\pi/2$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.